

PROMPT FOR RESOLVING CONWAY'S 99-GRAPH PROBLEM

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ABSTRACT. This document contains a full frontier-model research prompt for deciding whether a strongly regular graph with parameters $(99, 14, 1, 2)$ exists. A positive answer is a finite adjacency-matrix certificate; a negative answer requires a complete structural or certified exhaustive proof. Status snapshot: 20 July 2026.

1. Prompt

Current task statement

A strongly regular graph with parameters (v, k, λ, μ) is a simple graph on v vertices in which every vertex has degree k , every adjacent pair has exactly λ common neighbors, and every nonadjacent pair has exactly μ common neighbors.

Resolve Conway's 99-graph problem completely:

Does an $\text{srg}(99, 14, 1, 2)$ exist?

Equivalently, decide whether there is a symmetric 99×99 zero-one matrix A with zero diagonal such that

$$A\mathbf{1} = 14\mathbf{1}, \quad A^2 = 12I - A + 2J.$$

Assume for purposes of this task that a complete resolution is reachable. Do not assume which answer is correct.

Complete-resolution standard

A complete solution must prove exactly one alternative:

- Positive alternative: give a simple graph on 99 vertices satisfying all four parameters exactly, together with an exact independent verifier;
- Negative alternative: prove that no such graph exists, with a complete structural argument or a certified exhaustive search covering every isomorphism class;
- in either case, prove that all imposed normalizations, local configurations, automorphism assumptions, and symmetry-breaking constraints preserve the full existence question.

What does not count

Partial progress is insufficient unless it logically implies the exact resolution above. In particular, the following do not count:

- a graph matching only the degree and spectrum but not all common-neighbor conditions;
- a partial incidence structure, local neighborhood, or equitable partition that has not been completed to 99 vertices;
- an infeasible SAT instance without a checkable certificate and a proof that its encoding is equivalent to the graph problem;
- excluding graphs with a particular automorphism group without proving the remaining automorphism-free cases;
- a spectral, modular, or feasibility condition that is necessary but not sufficient;
- a numerical heuristic, failed search, or candidate adjacency matrix with even one parameter violation.

Use multiagent v2 aggressively and dynamically. You have up to 64 concurrent agents available. Do not use a fixed assignment such as "N agents for strategy X." Instead, manage the search using the following heuristics:

- Begin with a genuinely diverse portfolio of approaches. Agents should explore substantially different formulations, invariants, reductions, algebraic viewpoints, structural inductions, decompositions, optimization methods, exact computational searches, and formal-proof routes.
- Do not tell most agents the currently favored approach. Preserve independence during early rounds so that agents do not all converge to the same attractive but incomplete reduction.
- Maintain an explicit registry of approach families. Group agents by the mathematical mechanism they are using, not by superficial wording. If too many agents converge to one family, redirect some toward underexplored formulations.
- Do not allow one route to dominate merely because it gives elegant reductions. A route that ends at a missing lemma equivalent in strength to the original problem is not close to completion unless it supplies a genuinely new proof of that lemma.
- When an approach stalls at a theorem-strength missing lemma, mark that route as blocked. Reopen it only when an agent proposes a materially new mechanism, invariant, construction, or exact experiment.
- Keep several incompatible proof or construction routes alive through multiple rounds. Cross-pollinate ideas only after independent agents have developed them far enough to expose their real strengths and gaps.
- Use adversarial agents throughout. Every candidate theorem, construction, certificate, reduction, and program must be attacked independently. Assign separate agents to mathematical correctness, completeness of case analysis, numerical exactness, implementation soundness, and novelty checking.
- Require agents to return concrete lemmas, constructions, equations, counterexamples to proposed sublemmas, executable encodings, or formally stated proof obligations. Reject status reports, vague optimism, and claims that a global compatibility step is "routine."
- The root agent should repeatedly synthesize, challenge, redirect, and launch new rounds. Do not stop after the first wave fails. Convert promising patterns into exact conjectures and send those conjectures to independent proof and falsification teams.

Problem-specific search portfolio

In the initial independent rounds, ensure that the portfolio includes, but is not limited to, the following genuinely different families:

- fix a vertex and exploit that its 14-vertex neighborhood is a disjoint union of seven edges, then classify the second subconstituent and attachment patterns;
- use the Bose-Mesner algebra, eigenvalues $14, 3, -4$ with their forced multiplicities, star complements, interlacing, and integral-module constraints;
- derive and solve triple-intersection, quadruple-intersection, Krein, absolute-bound, and local-configuration constraints;
- encode the adjacency matrix or a reduced local completion as SAT, SMT, CP-SAT, exact cover, or integer programming with proof-producing solvers;
- use canonical augmentation and graph isomorphism to enumerate local neighborhoods, partial geometries, or coherent configurations without duplication;
- investigate links with partial geometries, block designs, two-distance sets, spherical embeddings, ternary or binary codes, and switching constructions;
- classify possible automorphism groups but keep a fully general automorphism-free branch alive;
- search explicitly for Cayley, voltage, lift, orbit, block-circulant, and algebraically generated candidates while independently pursuing nonexistence.

Mandatory adversarial audit

Any surviving candidate must be checked against every item below by agents that did not originate the candidate:

- check that A is exactly 99×99 , symmetric, binary, and has zero diagonal;
- check all row sums are 14 and verify $A^2 = 12I - A + 2J$ entry by entry in exact integer arithmetic;
- confirm adjacent pairs have exactly one common neighbor and nonadjacent pairs exactly two, without

relying only on a spectrum calculation;

- validate the forced spectrum and multiplicities as a redundant consistency check;
- audit every local-case enumeration for isomorph completeness and every symmetry-breaking clause for equisatisfiability;
- if nonexistence is computational, independently regenerate the case split and validate every unsatisfiability proof with a trusted checker;
- ensure no multigraph, loop, directed edge, duplicate vertex, or mislabeled incidence structure has been substituted for the required simple graph.

Required final artifact

The final response must include a conventional proof and the complete finite certificate appropriate to the direction:

- for existence, the adjacency matrix and edge list with hashes, plus at least two independent exact verifiers;
- for nonexistence, the complete structural reduction, case manifest, proof-producing solver output, certificate checker, and source code;
- a machine-readable statement of the matrix equations and a human-readable explanation of why they are equivalent to $\text{srg}(99,14,1,2)$;
- reproducibility instructions specifying software versions, commands, expected hashes, and expected verification output.

Public search may be used for standard strongly regular graph theory, published parameter restrictions, and existing SAT formulations. Do not search for a purported private answer to the 99-graph problem or merely report that the parameter set is unresolved.

Do not return merely because current approaches fail or agents report theorem-strength gaps. Continue launching new rounds, reopening blocked approaches only when there is a genuinely new mechanism, and searching for fresh formulations.

Return only when a complete resolution has been found and survives adversarial audit. Do not return a reduction, a special case, an isolated missing lemma, an improved bound that leaves the target unresolved, a "best effort" literature review, or an explanation of why the problem is difficult.

Spend at least 8 hours of sustained search before even considering return or termination. If the execution environment imposes a hard stop before a complete resolution, do not fabricate success or elapsed work: preserve all machine-checkable artifacts and report only the strongest rigorously verified derivation together with the exact remaining gap.